

Project 2: Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

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OUTLINE

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PROBLEM STATEMENT

1. Recognize deepfake ✓
2. Images are shared online via internet or re-digitalized causing a compression and loss of details

STATE OF ART

DATASETS

- Datasets for deepfake detection contain everyday scenarios, but common deepfakes are more critical images, such as war and disasters
- Datasets contain only original images

MODELS

- Spatial methods search for blending artifacts or any other error in the image
- Frequency methods use Fourier transform or DCT to find artifacts in frequency spectrum
- Images are compressed

PROPOSED METHOD

- Skeleton of the model: ViT-B/16
 - unfreeze only last 3 layers to learn complex concepts
 - use only one net with 768 hidden features
- Complete both tasks at the same time through a shared backbone with a two head structure
- Ablation study: testing different weights for each task
 - $\text{total loss} = \alpha \text{Loss}_{\text{rf}} + (1 - \alpha) \text{Loss}_{\text{trans}}$

PROPOSED METHOD

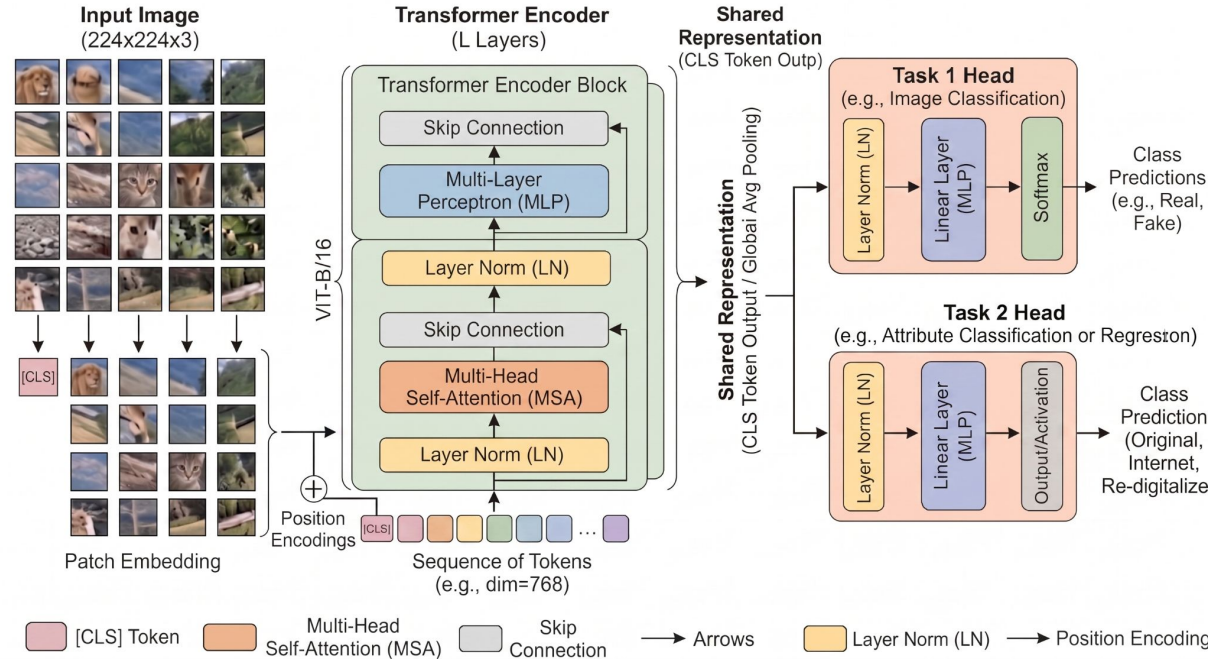


Figure 1. Multi-Task architecture based on ViT shared backbone and two independent heads.
Generated with Nano Banana 2.

DATASET



Figure 2. Real-Robust Bench overview from “Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios” (Li et al. 2025)

DATASET

Problem

[RRDataset_original_train_val.tar.gz](#) is **unbalanced**. It contains only original images, preventing the model to learn how to distinguish internet or re-digitized images

Solution

Use **only** [RRDataset_test.tar.gz](#) and divide it into train, evaluation and test sets:

- 70% train set
- 15% evaluation set
- 15% test set

Images are resized and centered (224x224) to standardize the inputs. Images in train set are randomly flipped horizontally and rotated to prevent overfitting

<https://zenodo.org/records/14963880>

DATASET: TRAIN DATASET ISSUE

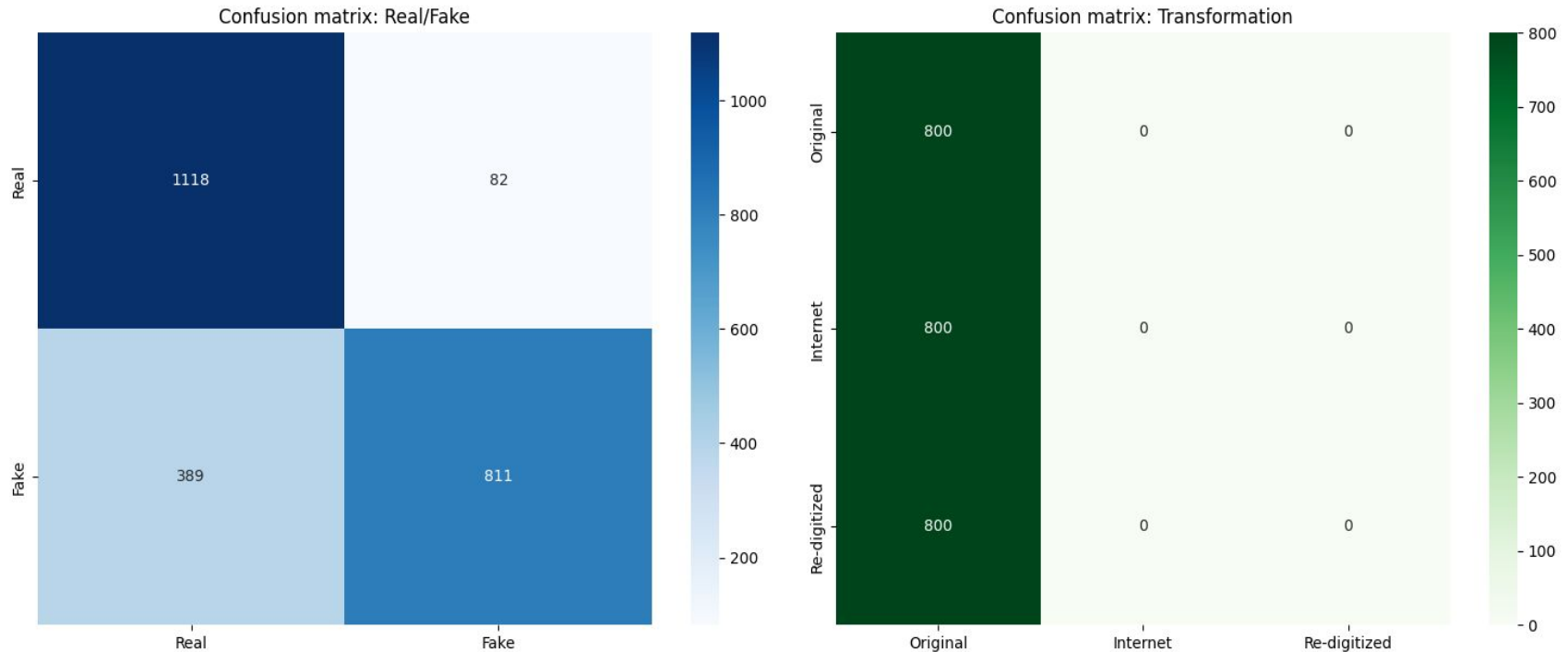


Figure 4. Output performances with RRDataset_original_train_val.tar.gz as training set.

DATASET BALANCE

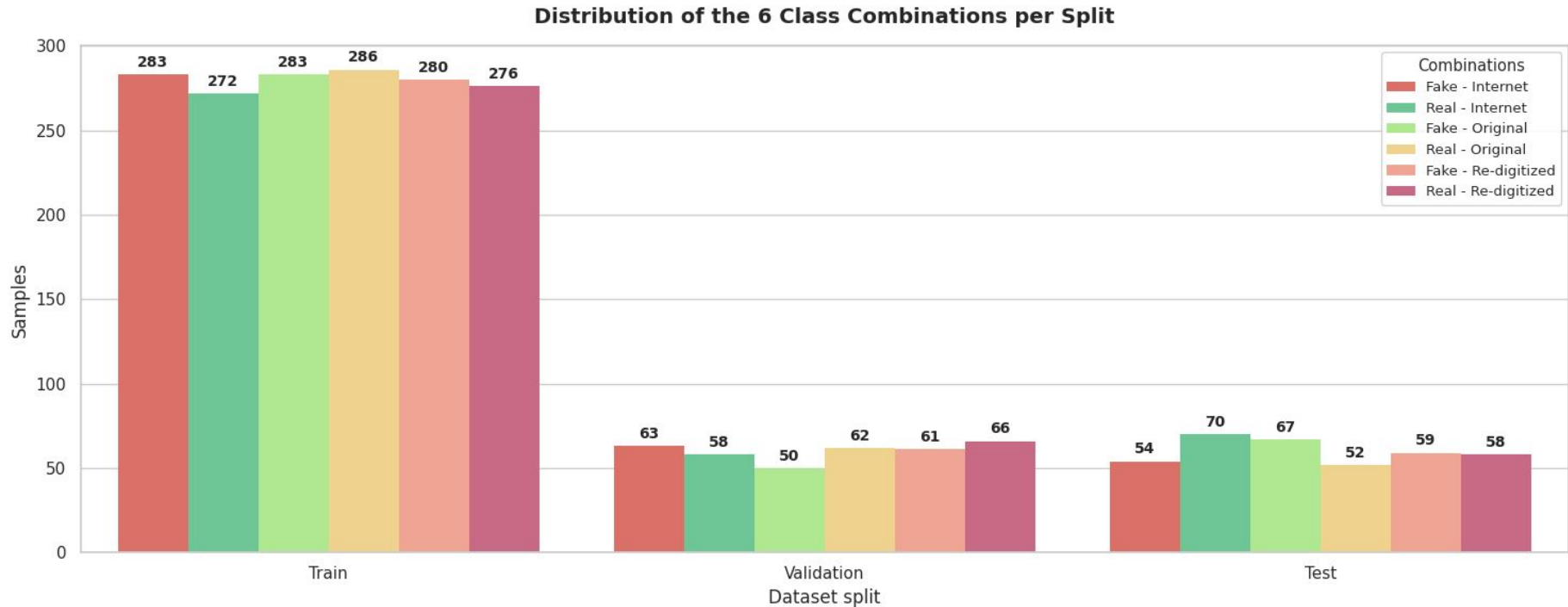


Figure 3. Distribution of classes and transformations in the three subsets (Train, Validation, Test).

EXPERIMENTAL SETUP

- SEED = 42 for reproducibility
- LEARNING_RATE = $1e-3$ with Adam optimizer
- EPOCHS = 20 (max) capped by early stopping
- PATIENCE = 3 epochs without improvement
- BATCH_SIZE = 64
- ALPHA $\in \{0.0, 0.5, 1.0\}$ for ablation study
- Dropout = 0.3

MODEL EVALUATION: ALPHA = 0.0

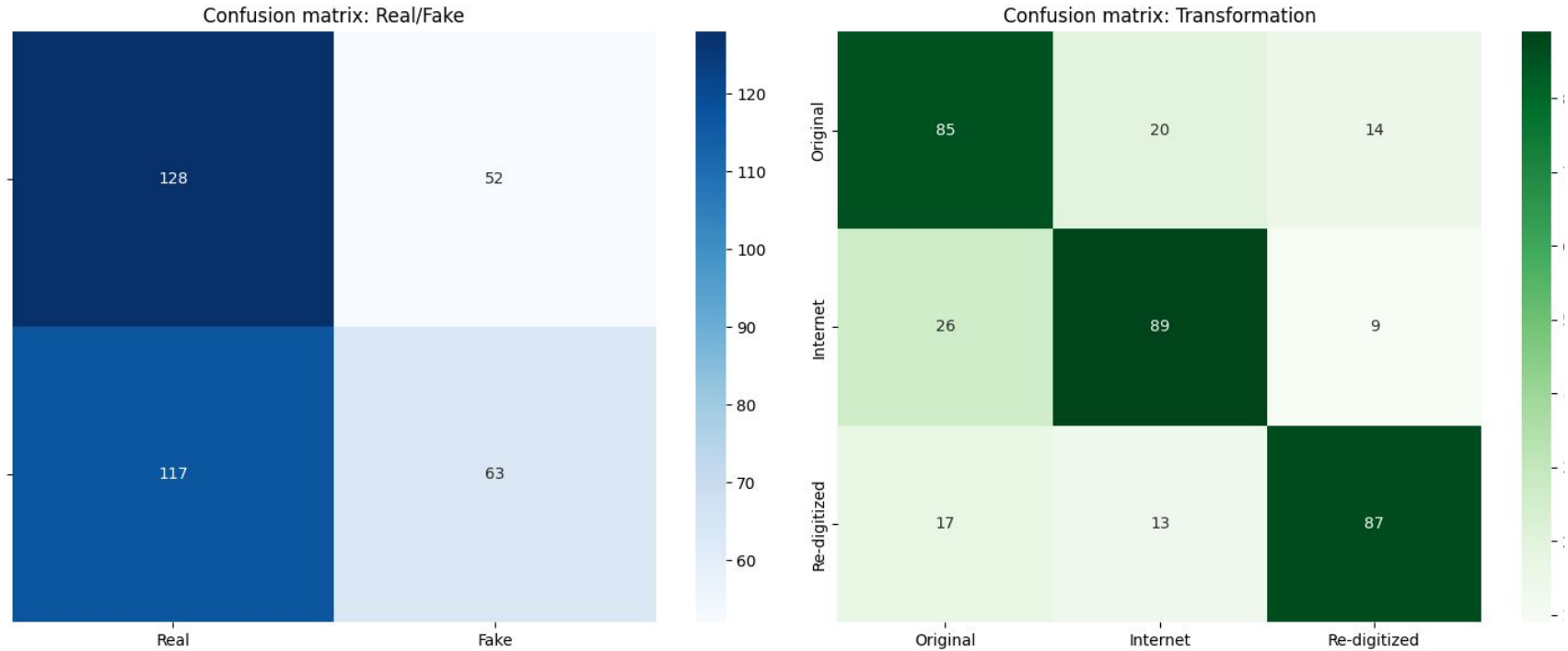


Figure 5. Output performances with ALPHA = 0.0

MODEL EVALUATION: ALPHA = 0.5

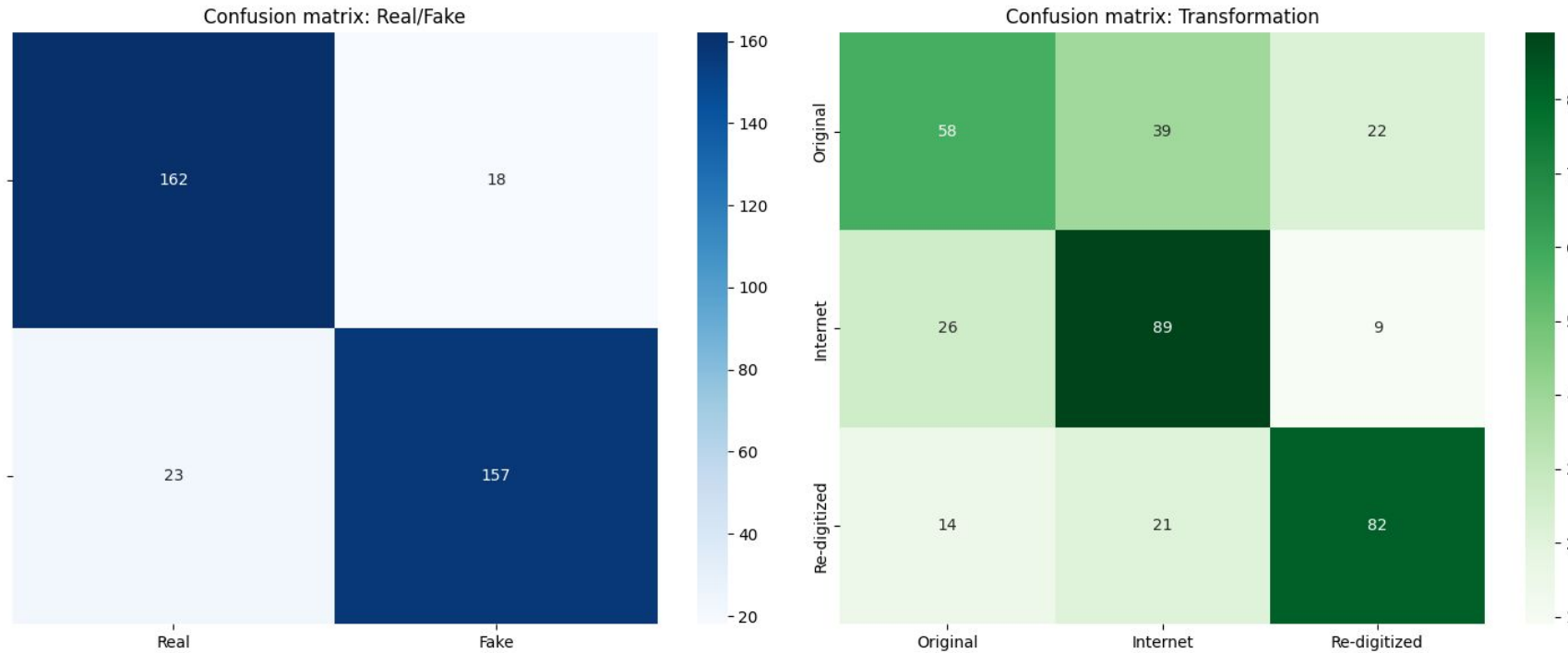


Figure 6. Output performances with ALPHA = 0.5

MODEL EVALUATION: ALPHA = 1.0

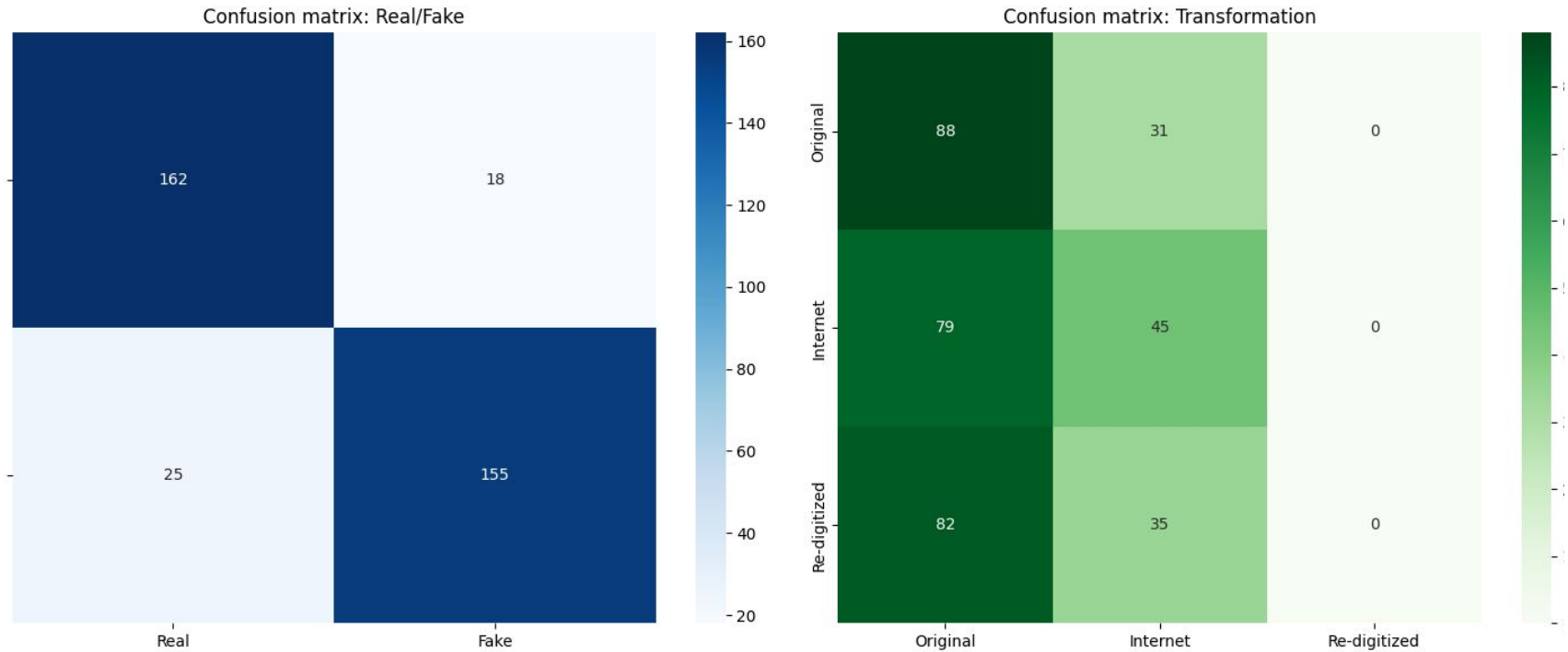


Figure 7. Output performances with ALPHA = 1.0

ABLATION STUDY

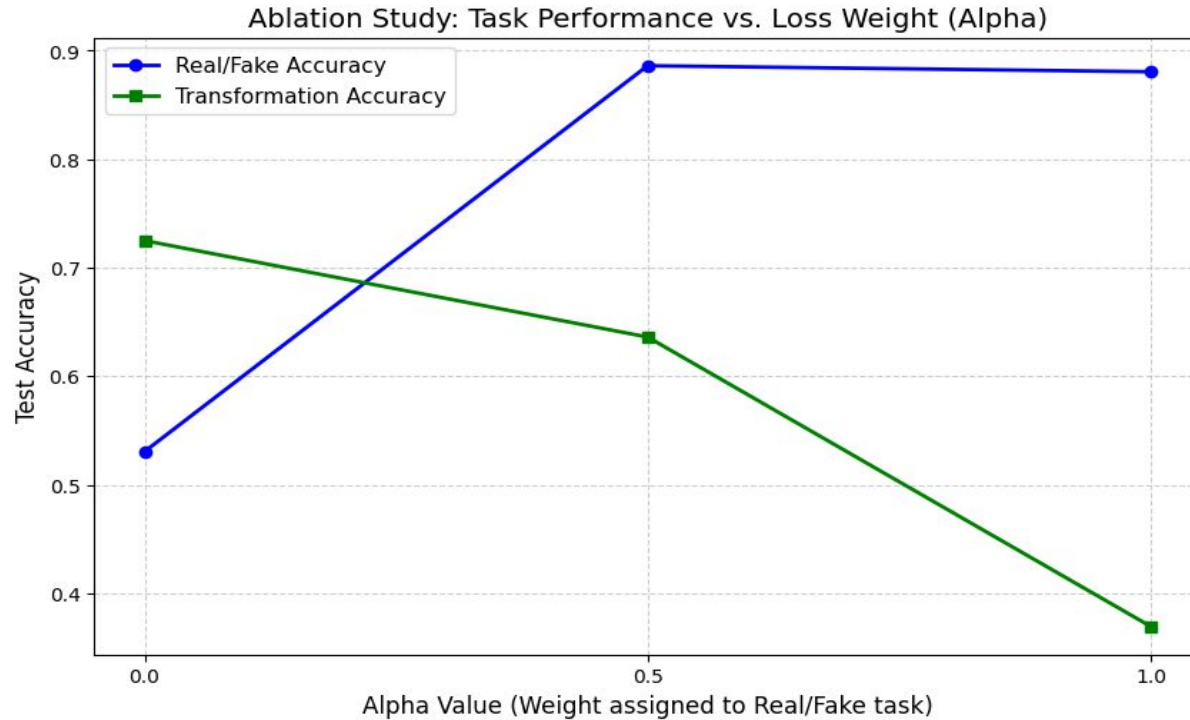


Figure 8. Ablation Study results: trend of Real/Fake and Transformation task accuracies as α varies.

REAL/FAKE ACCURACY PER TRANSFORMATION

α	Original Real/Fake	Internet Real/Fake	Re-digitized Real/Fake
0.0	88% / 18%	43% / 83%	90% / 10%
0.5	85% / 84%	93% / 81%	91% / 97%
1.0	88% / 85%	90% / 87%	91% / 86%

CONCLUSION

- Training exclusively on original images prevents models from operating in real-world scenarios
- The shared backbone extracts robust features capable of solving both Deepfake Detection and Transformation Classification simultaneously
- Internet compression remains the toughest challenge, as it destroys AI-generated artifacts

REFERENCES

Dataset:

- RRDataset - [[Dataset link](#)]

Literature:

- Li, Chunxiao, et al. "Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
- Shao, Rui, Tianxing Wu, and Ziwei Liu. "Detecting and grounding multi-modal media manipulation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023.

Course materials

- tutorial_2_cv_Vit.ipynb - [[Notebook link](#)]
- Medical Imaging.ipynb - [[Notebook link](#)]